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Practical Sheet

(01)

Web application is a program stored inside a web server and which can be accessed through a web browser without any installation. Users can log in using their accounts to use the service provided in the application without having to worry about versions of the software since it will always be the latest one.

(02)

Website is there to provide information to certain users while web application is there to provide a service to user.

(03)

The architecture of web sites primarily considered only a static webpage which provide necessary information to users while a web application architecture is mainly focused on user interactions and maintaining various services provided by them.

(04)

HTML, CSS, JS, SQL, Node.JS

(05)

Inspector in web browsers, text editors

(06)

Client side scripting is changing the interactive behavior of the web application when users interact with it, while server side scripting is used to change the interactive behavior between the server and the web application's front end.

(07)

Authentication is the process of verifying the users of a web application using given user details such as usernames and passwords.

(08)

The role of API is to maintain a connection between the client side and the server side of the web application when the user interact with the client side to make sure everything user sees is up-to-date and everything user does is saved in the server side.

(09)

First you should make a proper plan on how to build the application. Then teams should be divided according to client side and the server side and then the infrastructures required for the client side to communicate with the server side should be developed.

(10)

VSCode for IDE and Git for version control

(11)

React, Vue, Node, Express

(12)

Frameworks – larval, next.js, Django

CMS – shopify, wordpress

(13)

Front end development is the process of developing all the interfaces that the end user sees. It includes the web pages seen by the user such as login page, home page, etc

(14)

Back end development is the process of developing all the interfaces that the server uses to handle the workload of the web application. It is important since it is the one that actually does all the services provided by the web application

(15)

Websites can be made to handle scalability of them by developing the infrastructure that are used by the application according to a proper planning. This can help in handling larger amount of user interactions at a given time.

(16)

Security can be managed by using a good authorization methods and using methods to prevent unwanted access using techniques like SQL injections.

(17)

DBMS is used to handle and maintain all the user and application data in varies related tables which are stored in databases.

(18)

REST APIs are the connection between the front end and the back end of a web application. When a client request something from the application, its REST APIs job to go to the server to fetch that data and provide the users with them.

(19)

Validations of data requests, using encryptions

(20)

Responsive web design is designing a web site which can automatically adapt to any given screen size so users can access them using any device and still be able to get there work done without having to scroll in various directions to find the content.

(21)

Updates and maintenance are done regularly to the web applications stored in the servers and users can access the latest version of the application by simply refreshing the site which automatically loads the latest version of the application stored in the server.

(22)

Yes a static web site can transform into a web application by introducing backend servers, user authentications and various middleware architectures.

(23)

Web sites are more SEO friendly since they are premade HTML pages which can easily be loaded in the browser while application are progressively loaded

(24)

Web applications uses client side rendering where the work is done in the clients computer while websites use server side rendering where all the data is stored in the web server and only loaded to the client's computer when he is using them.

(25)

In websites the third party integration are handled by pre-installed packages in the browser using client's computer while web applications handle those by backend api calls to different providers so they are handled outside the client's computer.

(26)

Hyper text markup language

(27)

To introduce the title shown in the tab bar of the web browser

(28)

Content inside the head tag are not visible in the browser. It contains all the meta data and links to other related resources while content in the body is visible in the web page and it contains all the structural elements of the web page.

(29)

They are used to denote heading so readers can clearly identify them. The tags bolds the font and increase the font size of the content inside them.

(30)

P tag is used to display a formatted paragraph on the web page.

(31)

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Title</title>
```

```
  </head>
```

```
  <body>
```

```
    <!--some content -->
```

```
  </body>
```

```
</html>
```

(32)

